A Calibrated Separation of the Kontorovich–Lagarias and Volkov Exponent Predictions for the $5x+1$ Reverse Tree

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Abstract

Kontorovich and Lagarias predict a counting exponent $\eta \approx 0.650919$ for the $5x+1$ reverse tree, where $N_u(x) = x^{\eta+o(1)}$ for the accelerated map used here. A competing model due to Volkov predicts $\eta^* \approx 0.678$. The two authors write that Volkov's own data seems insufficient to discriminate between the values.

We enumerate the reverse tree exactly and extrapolate the truncation bias with Aitken's Δ^2. The fixed-window reading is 0.639; it excludes neither prediction. Applied to a branching process built to a known pressure-equation root, the same estimator under-reads by 0.038, more than the gap $\Delta = 0.027$ between the two disputed values, so a raw reading settles nothing.

Two more constructions share the arithmetic tree's branching law and, via an exact sibling congruence, its sibling spacing, drawing the branch residue instead of the true one: an integer recursion and its real-valued relaxation, both built to exponent 0.650919. Together with the branching process above, three constructions built to 0.650919 read within 0.0035 of that value, a band 0.0037 wide. The arithmetic tree reads 0.64926, below the two congruence-matched constructions and inside the wider band set by the third. A fourth construction, the same real-valued relaxation built to exponent 0.678, reads 0.0282 away, more than seven widths of that band outside it.

The comparison is a measurement, not a proof, calibrated against controls built to a known pressure-equation root. It targets the disputed exponent value; whether it bears on Volkov's branching model itself is a separate question.

1 Introduction

Kontorovich and Lagarias [1] study the $3x+1$ and $5x+1$ Collatz-type problems through the counting statistics of the associated reverse tree: for a fixed root u, the number $N_u(x)$ of integers up to x that eventually map to u under the accelerated map satisfies $N_u(x) = x^{\eta+o(1)}$ for a conjectural exponent η. Their stochastic model predicts $\eta_{5,BP} \approx 0.650919$ for $q = 5$ (their Theorem 8.10, subscript BP for branching process, in their own notation). A competing branching model due to Volkov [3], discussed in the same reference, grows a complete binary tree under a different encoding of the iterates and predicts $\eta^*_{5,BP} \approx 0.678$, a gap of $\Delta = 0.027$. Volkov's model counts a related but different quantity, $Q(x)$: non-divergent trajectories entering some cycle. Kontorovich and Lagarias note $\pi_5(x) \leq Q(x)$, where π_5 is their own un-accelerated counting function of §2's Remark, with the two orders of growth only conjectured to agree. Kontorovich and Lagarias write that "the empirical data Volkov presents seems insufficient to discriminate between these two predicted exponents," and separately note "some controversy concerning the conjectured value of the constant."

We measure the exponent by exact enumeration of the accelerated reverse tree. The fixed-window reading is 0.639, excluding neither prediction. Calibrated against constructions built to a known pressure-equation root, the arithmetic tree's reading falls inside the band set by three constructions built to 0.650919, and 0.0282 from a construction built to 0.678, more than seven band-widths away.

2 Setup: the reverse tree of the accelerated $qx + 1$ map

Fix an odd integer $q \geq 3$. The *accelerated $qx + 1$ map* is

$$T_q(n) = \frac{qn+1}{2^{v_2(qn+1)}}, \qquad n \text{ odd},$$

where v_2 is the 2-adic valuation. The *reverse tree* rooted at an odd integer u coprime to q is built by, at each odd node v, adjoining a child $w_a := (2^a v - 1)/q$ for every $a \geq 1$ such that $2^a v \equiv 1 \pmod{q}$; such w_a is automatically an odd integer whenever it is an integer. Writing $d := \operatorname{ord}_q(2)$, admissibility of a for a given parent v depends only on v mod q: if some $a_0(v) \in \{1, \dots, d\}$ satisfies $2^{a_0(v)} v \equiv 1 \pmod{q}$, it is unique, and the admissible exponents are exactly $a \equiv a_0(v) \pmod{d}$; nodes whose residue admits no such a_0 are sterile. At $q = 5$, $d = 4$, every nonzero residue admits an a_0, and only $v \equiv 0 \pmod{5}$ is sterile.

Every node has exactly one parent under T_q, so a given integer reaches u, if at all, along a single path. We count them:

$$N_u(x) := \#\{n \leq x : n \text{ odd and } T_q^k(n) = u \text{ for some } k \geq 0\}, \qquad N_u(x) = x^{\eta + o(1)}.$$

Remark 2.1. Kontorovich and Lagarias's Theorem 8.10 is a statement about the branching random walk $B[5^0]$: almost surely, its progeny count $I^*(x;\omega)$ satisfies $I^*(x;\omega) = x^{\eta_{5,BP}+o(1)}$. They use I^* as a proxy for the integer count $\pi_{u,5}(x)$, counting every integer reaching u under the un-accelerated map $n \mapsto (qn+1)/2$ for n odd, $n/2$ for n even. That $\pi_{u,5}$ shares $B[5^0]$'s exponent is their unproven heuristic. Separately, collapsing the 2-adic halving chain between consecutive odd values under $\pi_{u,5}$ changes its count by a factor at most $1 + \log_2 x$, which is $x^{o(1)}$, so $N_u(x)$ above shares $\pi_{u,5}(x)$'s exponent η whenever the latter exists. That η exists and does not depend on u is itself their Conjecture 7.2.

Lemma 2.2 (Sibling congruence)**.** *Let $q \geq 3$ be odd, $d = \operatorname{ord}_q(2)$, and v odd with $2^a v \equiv 1 \pmod{q}$ solvable. Write the children of v, in order of increasing admissible exponent, as $w_{(j)} := w_{a_0+jd} = (2^{a_0+jd}v - 1)/q$ for $j \geq 0$, where $a_0 = a_0(v)$. Then*

$$w_{(j+1)} = 2^d w_{(j)} + \frac{2^d - 1}{q} \qquad \textit{(exact identity in } \mathbb{Z}\textit{)},$$

and consequently $w_{(j+1)} \equiv w_{(j)} + c \pmod{q}$, with $c := \big((2^d - 1)/q\big) \bmod q$.

Proof. Direct substitution: $2^d w_{(j)} + (2^d - 1)/q = \big(2^{a_0+(j+1)d}v - 2^d + 2^d - 1\big)/q = w_{(j+1)}$. The congruence follows from $2^d \equiv 1 \pmod{q}$. $\square$

When q is prime, $c = 0$ would force $q^2 \mid 2^d - 1$, the base-2 Wieferich condition; the only known Wieferich primes are 1093 and 3511, so $c \neq 0$ for every q of interest here, and consecutive siblings' residues mod q advance by a fixed nonzero step. At $q = 5$, $d = 4$, $c = 3$. A real denominator q_{val} can replace q in two facts: dropping the additive constant, negligible once $w_{(j)}$ is large, leaves consecutive siblings at multiplicative distance $w_{(j+1)} \approx 2^d w_{(j)}$, independent of q; and every parent-to-child step itself carries the denominator, $w_a \approx 2^a v/q$, also dropping the additive constant. The residue step c stays integer, mod the original q. §5 uses both facts to build a branching process with the arithmetic tree's sibling spacing but a tunable exponent.

3 Method

We enumerate the exact reverse tree of T_5 from 300 roots: odd integers coprime to 5, sampled uniformly from $[101, 10^4)$ and outside every cycle of T_5, so the fixed measurement window, 10^5 through 10^8, lies entirely beyond them. A single depth-first pass tracks the running path-maximum at every node, giving simultaneous counts at truncation buffers 10^9 through 10^{13}; this agrees, count for count, with an independent buffer-by-buffer implementation on five of the 300 roots. A confidence interval built from a single fixed buffer excludes every candidate exponent once the sampling standard error falls below the residual truncation bias; the path-maximum construction extrapolates the buffer dependence directly, without re-running the enumeration once per buffer. The pooled mean slope across the 300 roots rises with truncation, its increments shrinking geometrically:

$$0.60049\ (10^9) \to 0.62387\ (10^{10}) \to 0.63261\ (10^{11}) \to 0.63650\ (10^{12}) \to 0.63801\ (10^{13}),$$

by a factor of 0.37 to 0.45 per decade of truncation: the signature of an extrapolable truncation bias.

4 Result: a fixed-window estimator

Aitken's Δ^2 extrapolation of this sequence to infinite truncation gives

$$\hat{\eta}_5 = 0.639, \qquad 95\% \text{ bootstrap interval, fixed window} = [0.633,\ 0.645]. \tag{1}$$

The bootstrap interval measures variation between sampled roots after the truncation extrapolation; it does not include the remaining bias from the fixed measurement window, and so cannot exclude either asymptotic prediction. That remaining bias is visible directly in a diagnostic panel: on 40 of the 300 roots, at the deepest buffer 10^{13}, the per-decade slope across $10^4 \to 10^5$ through $10^7 \to 10^8$ is still rising at the last decade tested, $0.6021 \to 0.6296 \to 0.6432 \to 0.6460$, with no plateau.

Empirical Result 4.1 (Finite-window reading of the counting exponent)**.** *The fixed-window estimator of the $5x+1$ reverse-tree counting exponent, computed by exact enumeration with Aitken-Δ^2-extrapolated truncation correction, is 0.639 (bootstrap interval $[0.633, 0.645]$). The sequence of local slopes rises monotonically across the available decades, still increasing at the last one tested; §5 extrapolates a related per-decade statistic at far greater truncation depth. On its own, this experiment does not provide a calibrated confidence interval for the asymptotic exponent and does not exclude the Volkov prediction.*

5 A calibrated comparison

The estimator behind Empirical Result 4.1 is biased. An *unconstrained construction* shares the arithmetic tree's offspring law but draws an independent residue at every node; built so its annealed pressure root is the Kontorovich–Lagarias value 0.650919 [2], and read with the same window, truncation, and extrapolation as §4, it gives 0.6131. That is an under-read of 0.038, larger than the gap $\Delta = 0.027$ separating the two disputed values.

The bias falls with depth, and depends on how much a process fluctuates, so it can be calibrated out by comparing readings at a common depth, on a different statistic: not the window-based estimator above, but the single-decade slope $\log_{10}(N(10^{k+1})/N(10^k))$, Aitken-extrapolated over the truncation buffers above 10^{k+1}, read at a checkpoint decade $10^k \to 10^{k+1}$ deep enough for the bias to have mostly decayed. *Congruence-matched constructions* share the arithmetic tree's offspring law and, via Lemma 2.2, its sibling spacing, with a drawn residue at each parent: an integer recursion, built to exponent 0.650919 only, since it keeps q integer; and its *real-valued relaxation*, built to either target. Replacing the integer denominator q in the parent-to-child step $w_a \approx 2^a v/q$ by a tunable real value q_{val} moves the relaxation's exponent α (the smaller root, α_-) to any target solving $q_{\text{val}}^{\alpha} = q(2^\alpha - 1)$, with q itself held fixed at the arithmetic tree's own admissibility rule and sterility probability. Three congruence-matched constructions result: the integer recursion and its relaxation, both built to 0.650919, and the relaxation alone, built to 0.678. The unconstrained construction of the paragraph above, sharing only the offspring law and drawing an independent residue at every node, bounds how much of the calibration depends on the sibling congruence. Every construction below runs on one common set of 300 roots, sampled by the same rule as §3, at checkpoints 10^4 through 10^{10} and truncation buffers 10^9 through 10^{15}, and the arithmetic tree's reading is checked against them.

Empirical Result 5.1 (Calibrated separation from the Volkov exponent)**.** *At the checkpoint decade* $10^9 \to 10^{10}$*, the two constructions built to exponent* 0.650919 *that share the sibling congruence read* 0.65122 *(integer recursion) and* 0.64981 *(its real-valued relaxation), a band* 0.00141 *wide. The unconstrained construction built to the same exponent reads* 0.64751*,* 0.0034 *below target; including it widens the band to* 0.0037*. The arithmetic tree reads* 0.64926*, below both constructions built to* 0.650919 *and inside the wider band set by the unconstrained one. A construction built to exponent* 0.678 *reads* 0.67748*,* 0.0282 *away from the arithmetic reading, more than seven widths of that wider band. The same comparison at the shallower decade* $10^7 \to 10^8$*, on the same checkpoint grid, gives an arithmetic reading of* 0.64622 *against* 0.67200 *for the exponent-*0.678 *construction, a gap of* 0.0258*: the separation grows with depth. This measurement targets the exponent value* 0.678*; Volkov's branching model itself uses a different encoding of the iterates and is not implemented here.*

Pushed further on the arithmetic tree alone, to checkpoints 10^{12} and buffers 10^{17}, past the point where the stochastic constructions develop a heavy tail that stalls their own extrapolation, the slope approaches 0.650919: 0.6465 at $10^7 \to 10^8$, 0.6487 at $10^8 \to 10^9$, 0.6490 at $10^9 \to 10^{10}$, then 0.6506 and 0.6505 at $10^{10} \to 10^{11}$ and $10^{11} \to 10^{12}$, the last two within 0.0005 of the prediction. This deeper grid's 0.6490 at $10^9 \to 10^{10}$ is close to, but computed on different checkpoint and buffer values than, the 0.64926 of Empirical Result 5.1. The bootstrap bands on these five readings cover root resampling only; a separate truncation term is at most 0.002, falling to at most 0.0004 from $10^8 \to 10^9$ onward.

6 Discussion

Two systematics bound Empirical Result 5.1. The congruence-matched integer recursion and its real-valued relaxation, both built to exponent 0.650919, agree to 0.0035 over six seeds on the raw log-slope at the deepest buffer, bounding any implementation systematic between the two recursions at about 0.004, comparable to the wider band width itself. Separately, the truncation extrapolation of §5 is accurate to at most 0.002, falling to at most 0.0004 from $10^8 \to 10^9$ onward. Both bounds are comparable to, or well below, the 0.0037 wider band width used to read the comparison, and well below the 0.0282 separation itself. At the calibration depth itself, the construction built to exponent 0.678 under-reads its own target by 0.00052 and the one built to 0.650919 by 0.00111, smaller still.

The 0.038 bias of the opening measurement in §5 and the residual bias at the calibration depth, under 0.004 for every construction there, come from related log-slope estimators of the same family: one over the wider window at shallow depth, one over a single decade at much greater depth. Both the added depth and the narrower window contribute to the drop.

The comparison settles the narrower question of whether the arithmetic tree's counting exponent is closer to 0.650919 or to 0.678. It does not implement or test Volkov's branching model directly, which encodes the iterates differently from the matched constructions built here. Collapsing the chain of doublings in Kontorovich and Lagarias's $B[5^0]$ gives each exponent $a \geq 1$ an independent chance $1/5$ of a branch at $2^a v/5$: this accelerated form of their model shares the unconstrained construction's mean offspring intensity, hence the same annealed pressure function, and by the companion paper's first-moment argument the same upper bound on the exponent [2]. The two constructions differ in their joint structure: the unconstrained construction draws one residue class per node, where the collapsed $B[5^0]$ assigns each exponent independently. On the same 300 roots and grid, the gap between the unconstrained construction's bias (0.038) and the arithmetic tree's (0.013, presuming the disputed exponent is 0.650919) measures how much less the arithmetic tree fluctuates than a process sharing $B[5^0]$'s accelerated intensity, consistent with the arithmetic tree fluctuating less than every construction in §5 and reading below both constructions built to 0.650919 at the calibration depth. The deep arithmetic-only run sits systematically at or just below 0.650919, a pattern the data here cannot yet distinguish from a slowly converging extrapolation.

Data Availability Statement

Code and data reproducing every claim in this paper, with a README per subfolder naming the result it supports, are at `https://github.com/faculdade/collatz-kl-volkov`.

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References

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